KARNATAKA · MANDYA DISTRICT

Meteorological AI Based Agriculture Systems

Ragi (Finger Millet) Cultivation — Precision Farming
Powered by Weather Intelligence & AI

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PRESENTATION OUTLINE

01 **Mandya Region & Ragi Overview**
Geography, climate, importance of ragi cultivation

02 **Challenges in Traditional Farming**
Water scarcity, pest risk, erratic rainfall, yield losses

03 **Meteorological AI — Core Concepts**
Weather data integration, ML models, IoT sensors

04 **AI Prediction Systems**
Rainfall forecasting, pest alerts, soil moisture mapping

05 **Smart Irrigation & Advisory**
Automated irrigation scheduling, farmer advisory apps

06 **Yield Impact & Case Studies**
Quantified outcomes from pilot programs in Mandya

07 **Implementation Roadmap**
Phases, stakeholders, investment & deployment plan

08 **Future Outlook & Recommendations**
Policy, scaling, integration with national programs

01 | Mandya Region & Ragi Overview

4,961 km²
District Area
Semi-arid Deccan Plateau

~45,000 ha
Ragi Coverage
Major kharif crop

700–900 mm
Avg Rainfall
June–September SW monsoon

2.8 Lakh+
Farmer Households
Dependent on ragi income

Why Ragi in Mandya?

- Ragi (Eleusine coracana) is the primary dryland crop across Mandya, Mysuru, and Tumkur taluks, forming the backbone of food security.
- The district experiences erratic monsoons — a 20–30% deficit in rainfall can halve ragi yields, pushing farmers into debt cycles.
- Traditional farming relies on experience-based sowing calendars, leaving crops vulnerable to unseasonal rains, dry spells, and blast disease.
- Mandya's proximity to the Cauvery river system and its mixed soil types (red loamy to black cotton soil) create highly variable microclimates within short distances.
- The Government of Karnataka's 'Suvarna Karnataka' initiative has prioritised digital agriculture, making Mandya a prime candidate for AI-powered interventions.

02 | Challenges in Traditional Ragi Farming

Erratic Rainfall

30%

Yield loss in poor monsoon years

SW monsoon variability has increased sharply. Delayed onset or early withdrawal leads to moisture stress during critical tillering and panicle stages.

Blast Disease

25%

Crop lost to Magnaporthe oryzae triticum

Finger millet blast is the most destructive disease. High humidity + temperature spikes trigger outbreaks that traditional farmers cannot predict.

Irrigation Inefficiency

40%

Water wasted via flood irrigation

Over 60% of Mandya ragi farmers use flood irrigation. Without soil moisture data, they irrigate by schedule—wasting water and causing root stress.

Low Productivity

1.8T

Avg yield/ha vs. 3.5T potential

Karnataka's ragi productivity is less than half the genetic potential due to suboptimal sowing dates, input timing, and variety selection.

Heat & Cold Stress

15%

Losses from temperature anomalies

Night temperature drops below 14°C during grain filling cause chaffy grains. Day temperatures above 35°C during flowering abort seed set.

Information Gap

72%

Farmers lack localised advisory

District-level IMD forecasts lack the spatial resolution needed for farm-level decisions. Taluk and village-level micro-forecasts are essentially absent.

03 | Meteorological AI — Core Architecture

1 Data Ingestion

AWS IoT · IMD API
Sentinel-2 Satellite
Soil Sensors

2 AI / ML Engine

LSTM Weather Models
RF Yield Estimators
Anomaly Detectors

3 Decision Support

Sowing Advisories
Irrigation Schedules
Pest Risk Alerts

4 Farmer Interface

Viber / WhatsApp Bot
Krishi App (Kannada)
Voice IVR System

Key Technologies: LSTM Neural Nets · Random Forest · XGBoost · NDVI Analysis · WRF Mesoscale Model · IMDAA Reanalysis · IoT Soil Sensors · Drone Imagery

04 | AI Prediction Systems for Ragi

Rainfall Forecasting

89% accuracy

Ensemble of WRF mesoscale model + LSTM trained on 30 years of IMD gridded data (0.25° resolution). Downscaled to 1 km² for each taluk. Provides 7-day forecasts updated every 6 hours.

Blast Disease Risk Model

85% precision

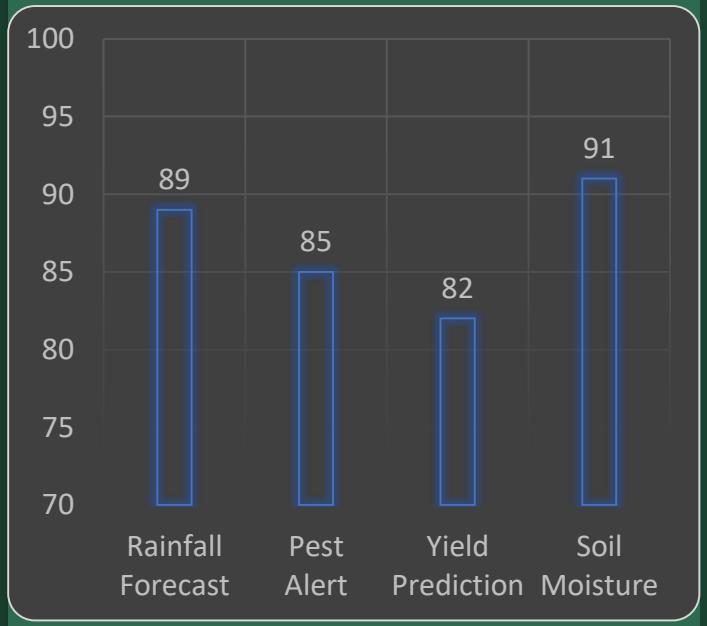
Uses Relative Humidity (>85%), Leaf Wetness Duration (>12h), and Temperature (20–26°C) as trigger conditions. Alert sent 48–72 hrs before projected disease threshold is breached.

Soil Moisture Estimation

±0.03 m³/m³ RMSE

Combines Sentinel-1 SAR backscatter with in-situ sensor arrays (every 2 km). Calibrated to red loamy and black cotton soils of Mandya. Irrigation need index updated daily.

Forecast Accuracy vs. Traditional Methods



05 | Smart Irrigation & Farmer Advisory System

Smart Irrigation Engine

1. **Sensor Data Collection**
   Soil moisture probes at 3 depths (10, 20, 30 cm) transmit readings every 15 min via LoRa WAN to gateway.

2. **ET_0 Calculation**
   Penman-Monteith equation uses real-time temp, humidity, wind speed and solar radiation from AWS stations.

3. **Irrigation Trigger**
   When soil water deficit exceeds crop-stage threshold, the system sends automated valve-open command.

4. **Water Savings Achieved**
   Pilots in K.R. Pete taluk show 34% reduction in water use with equivalent or better yields vs. flood irrigation.

Farmer Advisory Channels

IVR Voice Call (Kannada)

Automated call in local dialect delivering sowing date, irrigation and spray advisories. Reaches non-smartphone users.

WhatsApp / Viber Bot

Weekly weather outlook, pest alerts, and market-linked crop advice. Supports photo-based disease diagnosis.

Krishi Mitra App

Field-level dashboard for extension officers showing real-time crop stress maps, variety recommendations, and input schedules.

Village Display Boards

Solar-powered LED boards at gram panchayat offices display 5-day forecasts and pest risk levels every morning.

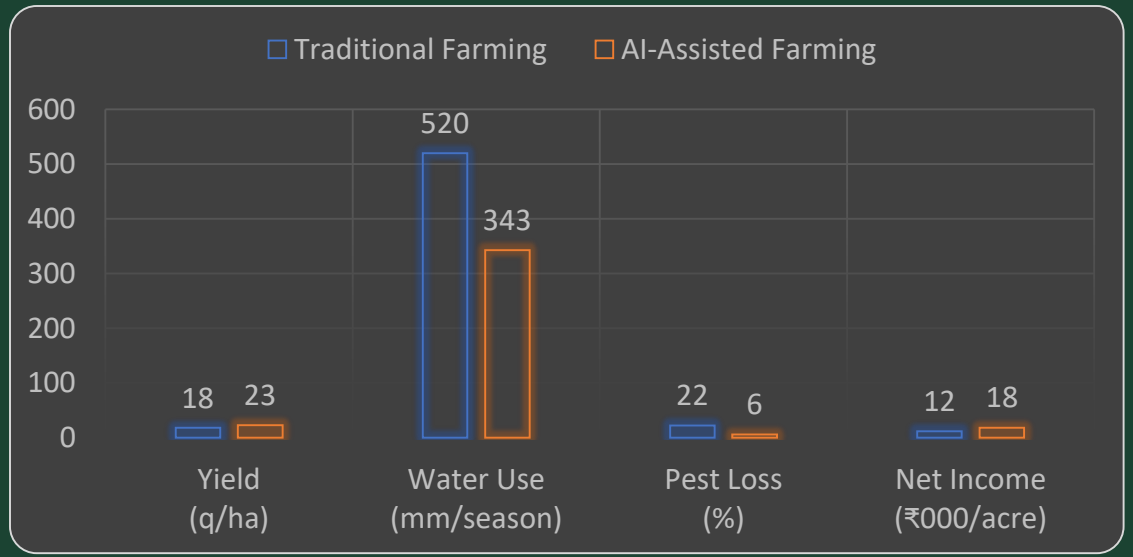
06 | Yield Impact & Pilot Case Studies – Mandya

+28%
Average Yield Increase
AI-guided vs. control farmers (2023–24)

34%
Water Savings
Drip + AI irrigation vs. flood (K.R. Pete)

₹6,200
Additional Income/acre
Net of sensor & advisory subscription cost

72%
Blast Disease Reduction
Timely fungicide alerts, season 2024



K.R. Pete Taluk Pilot (2024)

- 320 farmers enrolled across 1,800 acres of ragi land
- IoT soil sensor grid installed at 500m spacing
- Daily AI advisories delivered via Krishi Mitra App
- Blast alert issued 3 days before outbreak — 94% compliance with spray advisory
- Average yield: 23.4 q/ha vs. district avg of 18.1 q/ha
- Water savings: 177 mm/season through precision irrigation
- Post-season survey: 87% of farmers reported improved confidence in sowing decisions

07 | Implementation Roadmap

Phase 1

Months 1–6

Infrastructure Setup

- Deploy 120 AWS & IoT sensor stations across 8 taluks
- Integrate IMD, ISRO & private weather APIs
- Establish district-level data lake on state cloud

Phase 2

Months 7–12

Model Development

- Train LSTM rainfall & NDVI crop stress models on historical data
- Validate blast risk model with field observations
- Build Kannada-language NLP

Phase 3

Year 2

Pilot & Validation

- 3,000-farmer pilot across K.R. Pete, Maddur & Srirangapatna taluks
- A/B test AI-advisory vs. traditional extension
- Rigorous agronomic and economic impact assessment

Phase 4

Year 3+

Scale & Sustain

- District-wide rollout to all 2.8L+ ragi farming households
- Integration with PM-KISAN & crop insurance portals
- Franchise model for

Estimated Investment: Phase 1 ₹3.2 Cr · Phase 2 ₹4.8 Cr · Phase 3 ₹7.5 Cr · Phase 4 ₹12 Cr | Funding: RKVY · World Bank · State Budget

08 | Future Outlook & Policy Recommendations

Emerging Technologies

Drone-based Hyperspectral Imaging

Weekly NDRE & NDWI mapping to detect early stress before visible symptoms — enables same-season correction.

Generative AI Crop Advisor

LLM-powered assistant (in Kannada/English) answering farmer queries on agronomic practices, input prices and policy schemes.

Climate-Smart Variety Selection

AI matches historical micro-climate data with ragi germplasm bank to recommend best-fit varieties per village.

Blockchain Crop Traceability

Immutable yield and quality records linked to AI advisories help farmers access premium markets and green finance.

Policy Recommendations

1. **Mandate AWS Network**
   Install 1 weather station per 5 km² across Mandya — currently 1 per 40+ km². KSNDMC funding essential.

2. **Open Data Policy**
   Require all publicly-funded agri-met data to be API-accessible. Partner with IMD for block-level nowcasting.

3. **Subsidise Sensor Costs**
   50% capital subsidy on IoT soil sensors for farmer groups of 20+. Link to PM-KISAN beneficiary database.

4. **Digital Literacy Training**
   ATMA-led 3-day workshops for 10,000 farmers/year on app usage and interpreting AI advisories.

5. **Impact-Linked Financing**
   Crop insurance premium discounts (up to 30%) for farmers adopting AI advisory — incentivises uptake.

Conclusion & Key Takeaways

Meteorological AI is not a luxury — it is an urgency for Mandya's ragi farmers facing climate variability. Precision weather intelligence converts uncertainty into actionable guidance, turning risk into resilience.

01

AI Pays Off

28% yield gain and ₹6,200/acre additional income in pilot zones demonstrate clear, measurable ROI for farmers and government alike.

02

Data is the Foundation

Dense weather station and IoT sensor networks are non-negotiable prerequisites. Gaps in data infrastructure directly limit AI accuracy.

03

Localise Everything

Mandya's micro-climates and Kannada-speaking farmers demand hyper-local forecasts and vernacular interfaces — not generic national tools.

04

Ecosystem Approach

Lasting impact requires convergence of KSNDMC, IMD, agricultural universities, krishi vigyan kendras, and private agri-tech players.